

PlantECG

The First Polish Device for Measuring Photosynthetic Efficiency

An Advanced Portable Chlorophyll Fluorimeter

Plant Stress Meter

Plant Stethoscope for Assessing the Functioning of the Plant Heart (PSII)



Instructions prepared by:

Prof. Hazem M. Kalaji

Richard Grodowsky, MSc

Prof. Vasily Goltsev

Dr Wojciech Baba

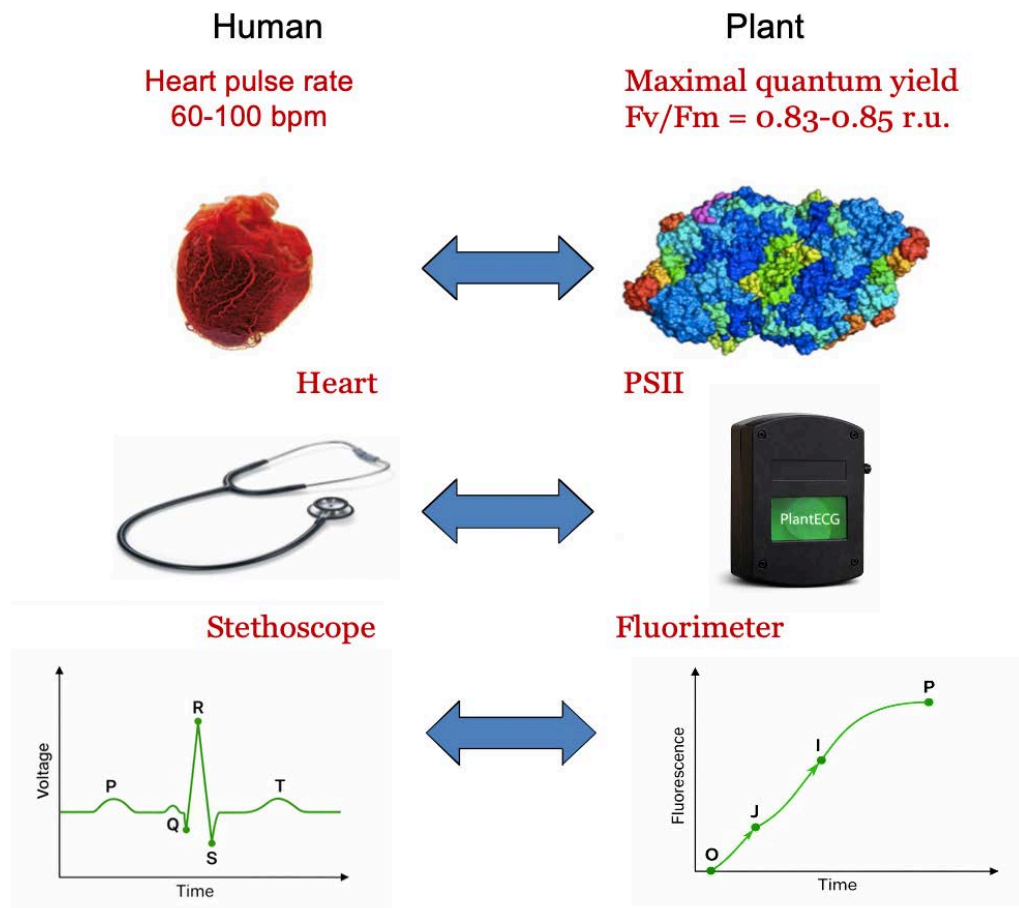
PlantECG is a fluorimeter, a breakthrough tool developed by Polish scientists, Prof. Hazem M. Kalaji and Ryszard Grodzowski, enabling early and non-invasive diagnosis of plant condition. It is the simplest, smallest, and most affordable system worldwide for recording chlorophyll fluorescence signals related to photosynthesis process, specifically the efficiency of Photosystem II (PSII) located in the thylakoid membranes of chloroplasts.



PlantECG is a fluorimeter that, in addition to recording chlorophyll fluorescence signals, is equipped with sensors for measuring ambient temperature, humidity, and atmospheric pressure, which are crucial for the proper interpretation of the obtained results. The device also features a visual LED indicator that informs about the physiological status of plants. Thanks to its unique technology, PlantECG enables ultra-fast measurements within one second, providing immediate information on the physiological condition and health of plants.

Prof. Hazem M. Kalaji is the only scientist in the world conducting research on photosynthesis based on an analogy between humans and plants, defining photosynthetic signals as the “beating heart of plants” or the plant electrocardiogram (Plant ECG). Prof. Kalaji believes that plants have their own language, referred to as **Plantish**, and he has demonstrated that communication with plants is possible through the analysis of photosynthesis, which is the most sensitive and informative physiological process in plant life.

The device has been designed to meet the needs of both professionals and beginner users, combining advanced technology with ease of use and intuitive interpretation of results. It is perfectly suited for scientists in fields such as agriculture, horticulture, forestry, biology, and environmental protection, as well as for practitioners including farmers. The device is also accessible to users outside the scientific domain, such as artists, architects, urban green space managers, and administrators responsible for parks, cities, and protected areas.



Why Plant ECG?

Just as an electrocardiogram (ECG) allows the assessment of human health, Plant ECG enables the recording of the “plant pulse”, that is, key physiological parameters reflecting plant vitality and functioning. This revolutionary approach opens a new era in plant monitoring and protection, strongly supporting the agriculture of the future.

Standard chlorophyll fluorescence measurements, such as the OJIP transient, F_v/F_m , PI_{abs} , and PI_{tot} , have long been applied in plant physiology research. However, adapting these well-established methods into a form analogous to a human ECG, as proposed by Prof. Hazem M. Kalaji, represents a fundamentally new concept characterized by the following innovations:

- A new diagnostic approach – instead of classical fluorescence curves, the results are interpreted in a manner analogous to electrocardiogram recordings, enabling immediate and intuitive assessment of the physiological status of plants.
- A unique technological solution – the developed PlantECG fluorimeter combines simplicity, miniaturization, and multifunctionality, including built-in sensors for temperature, humidity, and atmospheric pressure, together with ultra-fast measurements performed within one second.
- Practical applicability – by presenting results in the form of a “plant ECG”, the information becomes understandable not only to scientists, but also to farmers, gardeners, and decision-makers, thereby facilitating effective knowledge transfer and practical implementation in modern agriculture

- Originality and priority – to date, neither the scientific literature nor commercially available technologies have introduced the concept of a “plant ECG” in this form, making this approach original and pioneering.

Definition of new technology PlantECG:

PlantECG (Electrocardiogram of Plants) is an innovative diagnostic technology developed by Prof. Hazem M. Kalaji. It is based on the adaptation of chlorophyll fluorescence measurements into a form analogous to the recording of the human electrocardiogram. This technology enables ultra-fast, non-invasive assessment of plant physiological condition and the early detection of symptoms of both biotic and abiotic stress.

Elements of novelty and innovation:

- A new measurement concept – the transformation of chlorophyll fluorescence data into a “pulsating signal” of plants, reflecting the rhythm and functioning of the photosynthetic system.
- The first device of its kind – the PlantECG fluorimeter is the smallest and simplest device of this type worldwide. It integrates environmental sensors (temperature, humidity, and atmospheric pressure) together with a visual physiological status indicator (LED signals in green, yellow, and red), enabling immediate and intuitive interpretation of results.



- A new method of data interpretation – the simplified presentation of results, reminiscent of an ECG recording, allows intuitive analysis that is understandable both for scientists and for practitioners in agriculture and horticulture.
- Practical application – the technology supports precision agriculture, sustainable crop management, and early warning systems for environmental and stress-related factors.

Meaning and potential implementation:

- Scientific impact – introduces a new paradigm in research on photosynthetic signaling and stress diagnostics.
- Industrial impact – enables low-cost, rapid, and universal plant health assessment in agricultural farms, greenhouses, and laboratories.
- Socio-economic impact – facilitates effective knowledge transfer to the agri-food sector, increasing the resilience of food production systems to climate-related stress.

The most important features and functions:

- Non-invasive and non-contact measurement – performed without any damage to plants.
- Ultra-fast diagnostics – real-time results obtained within one second, with the ability to detect physiological changes in plants 2–3 days earlier than they become visible to the naked eye, enabling proactive sensing and early intervention.
- Versatile applications – suitable for ecological research (terrestrial and aquatic systems), agriculture, horticulture, and forestry.
- Assessment of physiological condition and plant health – precise analysis of Photosystem II (PSII) functioning.
- Early detection of plant stress (abiotic and biotic) – including drought, salinity, nutrient deficiencies, environmental pollutants, and biotic factors such as pathogen infections, fungal diseases, viral and bacterial infections, and other forms of biological stress affecting plant performance and vitality.
- Wireless data transmission – to computers and mobile devices.
- Environmental parameter monitoring – simultaneous measurement of temperature, atmospheric pressure, and relative humidity during each use.
- Alphanumeric display of measured parameters – clear and direct numerical readout.
- Visual plant health status indicator – color-coded LED system for rapid assessment (green, healthy plant; yellow-orange, moderate stress; red, severe stress).

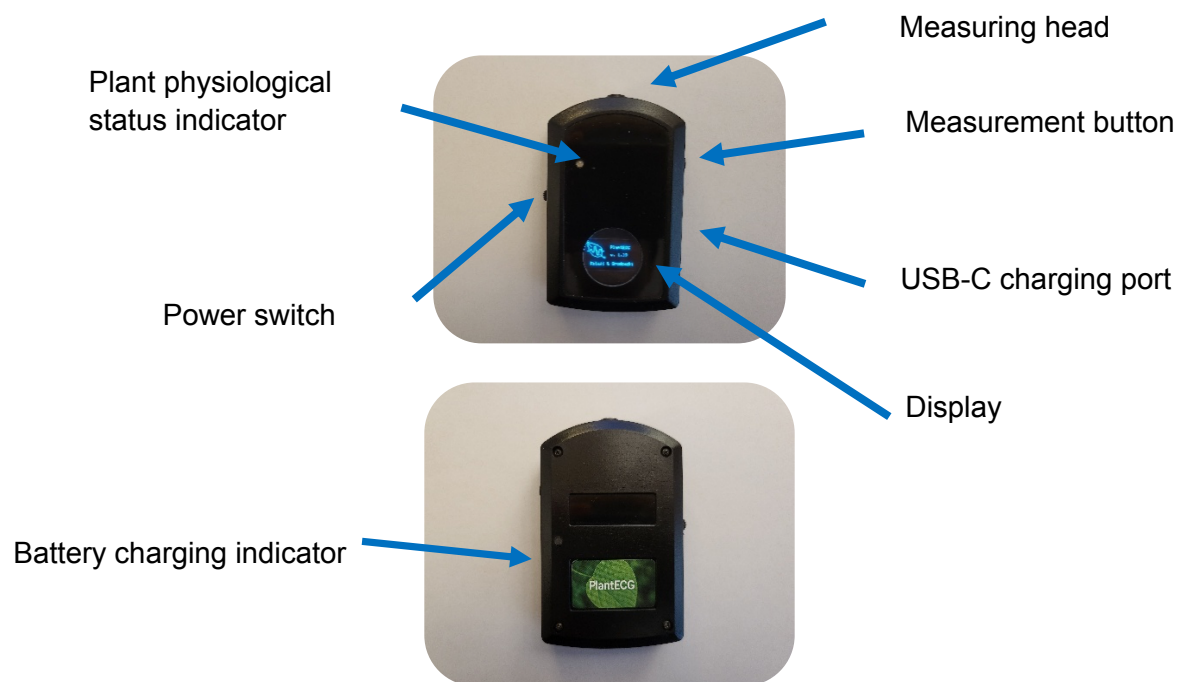
Technical Specifications:

- Type of measurement: non-invasive, with the possibility of measurement without dark adaptation of the plant/sample.
- Measurement time (fast chlorophyll fluorescence induction, OJIP): 1 second.
- Excitation light source: blue LED (430 nm).
- Excitation light intensity (illumination): approximately $4000 \mu\text{mol photons m}^{-2} \text{s}^{-1}$.

- Analyzed signal: activity of Photosystem II (PSII) and partial activity of Photosystem I (PSI).
- Operating mode: single measurement or series of recordings.
- Mobility: compact and lightweight design suitable for both field and laboratory use.
- Communication: connection to a computer, smartphone, tablet, or other devices via Wi-Fi; Wi-Fi access point 2.4 GHz (zero cables).
- Dimensions: 95 × 58 × 23 mm.
- Weight: 80 g.
- Detector: PIN photodiode with a 690 nm band-pass filter.
- Electronics: 32-bit controller with a 12-bit ADC converter.
- Battery: lithium-polymer, 3.7 V, 1200 mAh.

Charging: USB-C.

PlantECG description



The device consists of the following components:

1. Power switch
2. Measuring head
3. Measurement button
4. Display
5. Plant physiological status indicator
6. USB-C charging port
7. Battery charging indicator

Method of use

What do you need?

- PlantECG fluorimeter (main unit)
- Computer, tablet, or smartphone with an internet browser, enabling connection to the local Wi-Fi network generated by the fluorimeter (SSID: FLUO + the last two digits of the device ID, IP address: 192.168.35.1, password: 12345678)
- Plant samples for measurement
- Dark-adaptation chamber, darkened room, or light-shielding material (clip) to protect the sample from light (recommended). Green light may be used under dark conditions to facilitate easy measurements.

Device operation

Preparation:

The fluorimeter is battery powered. Before first use, charge the device via the USB-C port for approximately 12 hours. Full access to all functions is available after connecting an external device, such as a computer, tablet, or smartphone, via a web browser to the Wi-Fi network generated by the fluorimeter (Wi-Fi works like access point), IP address: 192.168.35.1, SSID: FLUO + the last two digits of the unique device identifier, password: 12345678.

Power switch:

Use the slider switch located on the side of the device. When the switch is moved toward the display, the battery is connected, allowing several hundred measurements after a full charge. When the switch is moved toward the measuring head, the battery is disconnected and the device is turned off. In this position, measurements can be performed only using an external power source, for example a power bank.

Charging indicator:

When the battery charge level is low, move the switch toward the display and connect the USB-C cable. The device can be charged directly from a computer or using a charger. During charging, the battery or charging indicator blinks. When the battery reaches a sufficient charge level, the indicator turns off.

Display and plant physiological status indicator:

Measurement results are shown on the display. The LED indicator reflects the plant physiological status based on the F_v/F_m value, default thresholds: green, $F_v/F_m > 0.75$, good condition; yellow, $0.35 < F_v/F_m < 0.75$, moderate stress; red, $F_v/F_m < 0.35$, severe stress.

The numerical threshold values can be defined in the configuration form at plantecg.pl/set. The same table includes the correction coefficient known as the “**Kalaji Factor**”, which should be experimentally adjusted within the range 1.00 to 1.08 depending on the plant species.

Performing a measurement:

Press the measurement button once to start a measurement. If the result is correct, press the button twice to add the fluorescence curve to the graph. Repeat the procedure to store multiple curves. Press the button twice again to end the measurement series.

Saving measurements:

After completing a measurement series, the data are automatically saved as a file named oijp.csv on the device connected via Wi-Fi.

Measurement procedure

Step 1: Dark adaptation

To obtain optimal measurement results, the leaf should be dark-adapted for 20–30 minutes before measurement. Dark adaptation restores the photosystems to their initial state and improves the reliability of the chlorophyll fluorescence signal. Use an appropriate clip (6 mm diameter), a dark-adaptation chamber (dark room), or cover the leaf surface with light-shielding material.

Step 2: Positioning the measuring head on the leaf surface

Gently press the measuring head against the leaf surface. Make sure that the measuring head fits tightly to the leaf surface to prevent lateral light penetration. Do not press the head too firmly, as this may damage the leaf tissue.

Step 3: Performing the measurement

Press the measurement button once to start the measurement. The device emits a light pulse and automatically records the chlorophyll fluorescence signals. The measurement takes approximately 1 second. During the measurement, do not move the measuring head or the leaf. After completion, the basic results are displayed on the screen, as shown in the image below.



Step 4: Saving the fluorescence curves displayed on the graph

If the graph displayed on the device connected to the fluorimeter is considered correct, press the measurement button twice to save it on the graph.

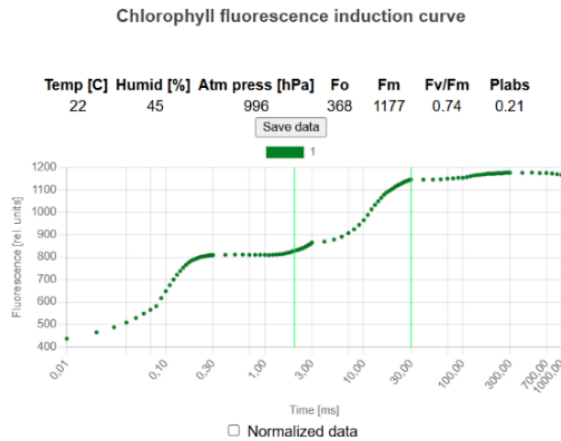
Step 5: Ending the measurement series

After saving the final measurement curve by pressing the measurement button twice, press the button twice again to end the measurement series. Previously recorded curves will be hidden, and the graph will display the mean value together with the range of variability (± 1 standard deviation).

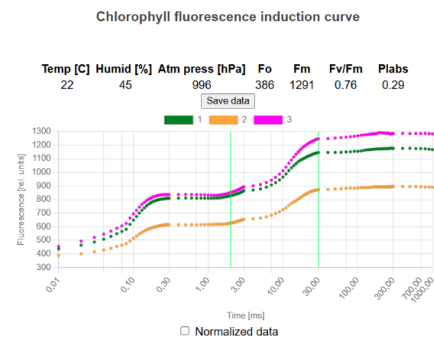
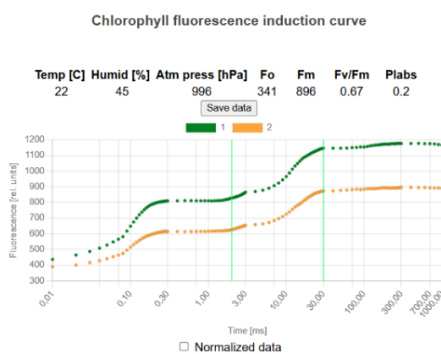
The measurement data are then saved as a file named ojip.csv in the non-volatile memory of the device connected to the fluorimeter.

Example of use:

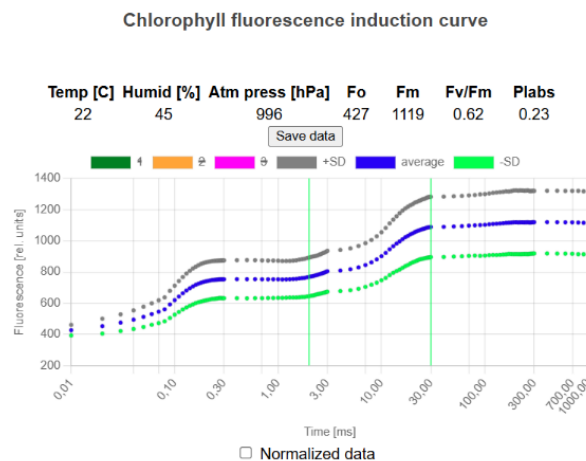
Press the measurement button once to perform the first measurement. The first curve will appear on the device screen:



Press the measurement button twice to save the first curve. Then press the measurement button once again to perform the second measurement. Press the button twice to save the second measurement. Perform the third measurement.



Press the measurement button twice to save the third curve, and then press it twice again. The screen will display the average value of the three saved measurements together with the range of variability ($\pm$ one standard deviation, $+SD / -SD$).



Press the measurement button three times or refresh the browser page to clear the screen. You can also refresh page from the browser directly. The device is then ready to perform the next measurement series.

Measurement principles and basic parameters (OJIP curve)

The device records the O–J–I–P chlorophyll fluorescence induction curve and analyzes the photosynthetic response across time scales ranging from microseconds to milliseconds (one point each 5 us):

O (Initial level): initial fluorescence when all reaction centers are open.

J (Step): phase of rapid fluorescence increase (1–2 ms).

I (Intermediate phase): phase of slower fluorescence increase (around 30 ms).

P (Peak): maximum fluorescence when all reaction centers are closed.

Basic parameters

Fo: minimum fluorescence intensity (reference value).

Fm: maximum fluorescence intensity.

Fv/Fm: maximum quantum efficiency of Photosystem II, typically ranging from 0.75 to 0.85 in healthy plants, defined as $(F_m - F_o) / F_m$.

PI(ABS): performance index indicating the efficiency of photochemical reactions.

Kalaji Factor: a correction factor applied to the calculation of the Fv/Fm parameter. This correction enables estimation of Fv/Fm values for plants measured without dark adaptation and displays results equivalent to those obtained after dark adaptation. The factor must be determined experimentally and stored in the fluorimeter memory (plantecg.pl/set). The default value is 1.08.

Analysis of measurement results

The recorded (displayed) chlorophyll fluorescence induction curves (118 data points) are saved in CSV format. These files can be imported into the PlantECG analyzer or a spreadsheet application for parameter calculation and graphical visualization.

CSV file format

- The first row of each file contains time values (in milliseconds).
- Subsequent rows contain fluorescence intensity values.
- Multiple measurement results can be stored in a single file.
- Environmental data recorded at the time of measurement, temperature, atmospheric pressure, and relative humidity, are stored at the end of the file.

Example:

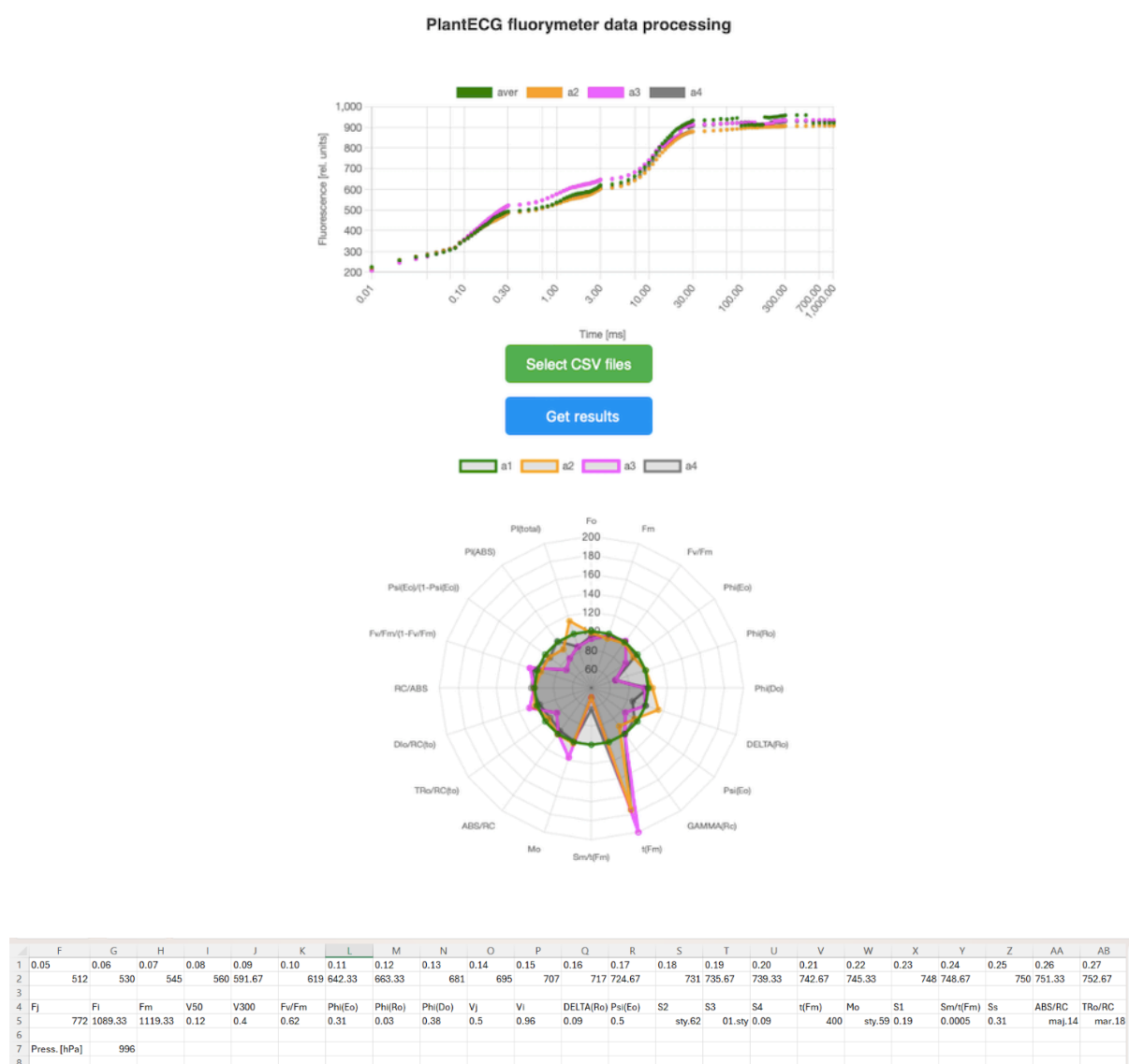
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	
1	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09	0.10	0.11	0.12	0.13	0.14	0.15	0.16	0.17	0.18	0.19	0.20	0.21	0.22	0.23	0.24
2	436	464	487	508	528	548	565	581	617	648	675	700	721	738	753	765	775	783	788	792	796	799	802	
3	389	400	416	428	441	454	464	475	495	515	531	546	558	568	576	584	589	593	597	600	603	606	609	
4	455	493	521	545	567	588	606	624	663	694	721	744	764	779	792	802	810	817	822	826	829	831	833	
5																								
6		Temp. [C]	22																					
7		Hum. [%]	45																					
8		Press. [hPa]	996																					
9																								
10		(C) Kalaji & Grodowski 2025																						

Analytical tool

The ojip.csv file saved during measurements can be opened in a spreadsheet application or uploaded to plantecg.pl using the option “Select CSV files”. The dedicated application automatically calculates the most important photosynthesis parameters based on OJIP analysis.

After clicking the “Download results” button, two analysis images and a results file (ojipres.csv) are downloaded to the device.

Example:



When opening CSV files in Microsoft Excel, select Data → From Text/CSV and set the field separator to a comma.

Parameters measured and calculated for Plant ECG (The comparison of the 3 sources)

The comparison below analyzes the definitions of the requested chlorophyll fluorescence parameters across all three sources:

Parameter	Definition/Formula (Kalaji et al., 2017)	Definition/Formula (Tsimilli-Michael, 2020)	Confirmation/Usage (Stirbet et al., 2025)
F_0	Initial (zero) fluorescence; minimal F when RCs are open.	Minimal reliable recorded F (at O-step), when all RCs are open.	F at O-step, minimal fluorescence when all RCs are open.
F_{50}, F_{100}, F_{300}	Fluorescence at specific microseconds ($F_{50\mu s}$ minimal, $F_{300\mu s}$ used in M_0).	Extracted data points; $F_{50\mu s}$ or $F_{20\mu s}$ is the minimal reliable F .	Extracted data points; $F_{300\mu s}$ is used in M_0 and related to the K-step.
F_J, F_I	F at Step J (2 ms) and Step I (30 ms).	F at J-step (2 ms) and I-step (30 ms) of OJIP.	F at J-step and I-step of OJIP transient.
F_M	Maximum F when all RCs are closed.	Maximal F , when all RCs are closed; equals F_P if actinic light is sufficient.	Maximal F when all RCs are closed; F_P equals F_M only if light intensity allows complete closure.
F_V/F_M (Φ_{P_0})	Maximum quantum yield of primary photochemical reactions; $\Phi_{P_0}=1-(F_0/F_M)$.	Maximum quantum yield for primary photochemistry, $\Phi_{P_0}=[1-(F_0/F_M)]$.	Maximum quantum yield of PSII primary photochemistry, Φ_{P_0} , derived as $1-(F_0/F_M)$.
Φ_{E_0}	Quantum yield of electron transport (at $t=0$); $\Phi_{E_0}=\varphi_{P_0}\cdot\psi_0$.	Quantum yield for electron transport (ET), $\Phi_{E_0}=[1-(F_0/F_M)]\times(1-V_J)$.	Quantum yield for ET, $\Phi_{E_0}\equiv ET_0/ABS=\Phi_{P_0}\times\psi_{E_0}$.
Φ_{R_0}	Not explicitly defined.	Quantum yield for reduction of PSI end electron acceptors (RE); $\Phi_{R_0}=[1-(F_0/F_M)]\times(1-V_I)$.	Quantum yield for RE, $\Phi_{R_0}\equiv RE_0/ABS=\Phi_{P_0}\times(1-V_I)$.
Φ_{D_0}	Quantum yield of energy dissipation (at $t=0$); $\Phi_{D_0}=(F_0/F_M)$.	Quantum yield of energy dissipation at time zero; $\Phi_{D_0}=F_0/F_M$.	Quantum yield of energy dissipation at time zero; $\Phi_{D_0}=1-\Phi_{P_0}=F_0/F_M$.
V_J, V_I	Relative variable fluorescence at Step J/I; $V_t=(F_t-F_0)/(F_M-F_0)$.	Relative variable fluorescence V_t applied at $t=J/I$ (normalization on F_M-F_0).	Relative variable fluorescence V_t applied at $t=J/I$.
δ_{R_0}	Not explicitly defined.	Efficiency/probability $\delta_{R_0}=(1-V_I)/(1-V_J)$ (electron transfer to PSI acceptors).	Efficiency/probability $\delta_{R_0}\equiv RE_0/ET_0=(1-V_I)/(1-V_J)$ (electron transfer efficiency from intersystem carriers to PSI end acceptors).
ψ_{E_0} (ψ_{E_0})	Probability that a trapped exciton moves an electron outside Q_{A-} ; $\psi_0=(1-V_J)$.	Efficiency/probability $\psi_{E_0}=(1-V_J)$ that an electron moves further than Q_{A-} .	Efficiency/probability $\psi_{E_0}\equiv ET_0/TR_0=(1-V_J)$ that an electron moves further than Q_{A-} .
S_m	Normalized total area above the OJIP curve (multiple turnover Q_A reductions); EC_0/RC .	Normalized Area $S_m\equiv Area/F_V$, measure of total electron carriers reduced per RC (EC_0/RC).	Normalized Area $S_m\equiv Area/F_V$, a measure of total electron carriers reduced per RC (EC_0/RC).
M_0	Approximated initial slope of $V=f(t)$; reflects net rate of PSII RCs closure.	Approximated initial slope $M_0\equiv 4\times(F_{300\mu s}-F_{50\mu s})/(F_M-F_0)$.	Approximated initial slope of $V_t=f(t)$ kinetics; $M_0=TR_0/RC-ET_0/RC$ (net rate of RC closure).
ABS/RC	Absorption flux per one active RC (apparent antenna size).	Absorption flux per RC, $M_0\times(1/V_J)\times(1/\Phi_{P_0})$ (measure of PSII apparent antenna size).	Absorption flux per RC; equivalent to $\Phi_{P_0-1}\times TR_0/RC$ (PSII apparent antenna size).
TR_0/RC	Energy flux trapped by one active RC at $t=0$; $M_0(1/V_J)$.	Trapped energy flux per RC; $M_0\times(1/V_J)$.	Trapped energy flux per RC; $M_0\times(1/V_J)$.
DI_0/RC	Dissipated energy flux per RC; $(ABS/RC)-(TR_0/RC)$.	Not explicitly listed in glossary.	Dissipated energy flux per RC; $ABS/RC-TR_0/RC$.
RC/ABS	Efficiency index/density of RCs per chlorophyll; reciprocal of ABS/RC .	RCs per PSII antenna Chl a ; reciprocal of ABS/RC .	RCs per PSII antenna Chl a ; reciprocal of ABS/RC , related to $\gamma_{RC}/(1-\gamma_{RC})$.

γ_{RC}	Not explicitly defined.	Probability that a PSII Chl <i>a</i> molecule functions as RC.	Probability that a PSII Chl <i>a</i> molecule functions as RC; $\gamma_{RC} = RC/(ABS+RC)$.
$\Phi_{Po}/(1-\Phi_{Po})$	Ratio of primary photochemical reaction efficiency; equals F_V/F_O .	Partial performance index related to $\log(PI)$.	Partial performance index term in PI_{ABS} calculation; equals k_P/k_N .
$\Psi_{Eo}/(1-\Psi_{Eo})$	Ratio describing the efficiency of electron transport outside Q_A^- .	Partial performance index related to $\log(PI)$.	Partial performance index term in PI_{ABS} calculation.
PI_{ABS}	PSII performance index on an absorption basis (energy conservation up to intersystem acceptors).	Performance index for energy conservation up to reduction of intersystem electron acceptors (3 terms).	Performance index for energy conservation up to reduction of intersystem electron acceptors (3 terms).
PI_{total}	Not explicitly defined.	Performance index for energy conservation up to reduction of PSI end electron acceptors (4 terms).	Performance index extended to include energy conservation up to reduction of PSI end electron acceptors (4 terms).
k_N, k_P	Rate constant of non-photochemical/photochemical reactions.	Nonphotochemical/photochemical deexcitation rate constants.	Nonphotochemical (k_N) and photochemical (k_P) deexcitation rate constants, derived from F_O and F_M .

References:

[1] A. Stirbet, G. Govindjee, and A. Srivastava, “In honor of Reto Jörg Strasser: A pioneer of chlorophyll a fluorecence research,” *Photosynthetica*, vol. 63, no. 3, p. 267, 2025.

[2] M. Tsimilli-Michael, “Revisiting JIP-test: An educative review on concepts, assumptions, approximations, definitions and terminology,” *Photosynthetica*, vol. 58, no. SI, pp. 275–292, 2020.

[3] M. H. Kalaji, V. N. Goltsev, K. Žuk-Gołaszewska, M. Zivcak, and M. Brestic, *Chlorophyll Fluorescence: Understanding Crop Performance—Basics and Applications*. CRC Press, 2017.

Good practice:

Measurement consistency: Due to light-induced photo-adaptation, measurements on the same plant should be performed on different areas of the leaves.

Time of day: Outside laboratory conditions, comparative measurements should be carried out at consistent times of day to minimize diurnal variation in environmental parameters such as sunlight, temperature, and humidity.

Replicates: Increasing the number of measurements on multiple leaves of the same plant improves the statistical reliability of the results.

Number of measurements: In experimental studies, at least 9 measurements per level of the tested variable are recommended. In field studies, due to higher environmental variability, a larger number of measurements is advised. Weather conditions and seasonal trends should also be taken into account.

Troubleshooting

Weak or unstable signal

If measured values are lower than expected or unstable, check the following:

- * Proper contact between the measuring head and the leaf surface
- * Cleanliness of the measuring head and the leaf surface
- * Sufficient dark-adaptation time
- * Adequate battery charge level, indicated by missing points on the curve or a step-like appearance of the curve
- * The clip is open (if you are using leaf clips for plant/sample dark adaption)

Device does not respond

- * Charge the device for several hours, at least 12 hours
- * Turn the device off and restart it
- * If the problem persists, contact technical support

Safety and maintenance

Light exposure: The device emits high-intensity light pulses. Never direct the measuring head toward the eyes.

Cleaning: Clean the measuring head with a soft, dry cloth. Do not use strong solvents or abrasive agents.

Storage: Store the device in a cool, dry place, away from direct sunlight.

Battery: Charge the battery regularly and consult service support if battery performance significantly deteriorates.

Contact and support

For assistance with measurements, device maintenance, or other technical support, please contact the manufacturer or an authorized distributor. Additional information and application notes are available at plantecg.pl.